The A-type divisors impose no condition:

the span of Motegi's type-C basis is *exactly* the A-type GKM lattice, and every structure constant is regular there for *every* unit renormalisation

closing gap 2 of `2026-10-17-c2` — and reporting that the pre-registered branch fired: gap 2 was never a gap, and the reason is a theorem

Clio

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Author	Clio
Date	18 October 2026
Recipient	Rick (`grandparick20@gmail.com`)
Repository	`clio-vega/proofs`
Commit	`c142ac5`

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Abstract

Let $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ be Motegi's $\mathcal{E}^C$-normalised type-C weight functions, indexed by subsets $\mathbf{M},\mathbf{I}\subseteq[N]$, regarded as the restriction of a basis $\{\phi_{\mathbf{I}}\}$ to the 2^N torus-fixed points of $LG(N,2N)$. Three families of divisors in w-space carry the three families of positive roots: the *pair* divisors $\{w_aw_b=1\}$ $(e_a{+}e_b)$, the *boundary* divisors $\{w_m^h=1\}$ $(h{\cdot}e_m)$, and the *A-type* divisors $\{w_i=w_j\}$ (e_i-e_j). `2026-10-17-c2` settled the first two and left the third as gap 2.

This paper settles the third, and the answer has a different shape from the other two. Write s_{ij} for the transposition acting on subsets and $\mathcal{O}_{ij}$ for the local ring at $\{w_i=w_j\}$.

Lemma A1. $\widetilde{A}_{\mathbf{M},\mathbf{I}}\equiv\widetilde{A}_{s_{ij}\mathbf{M},\mathbf{I}}\pmod{w_i-w_j}$, for all N, all $\mathbf{I}$, and *both* types B and C: the A-type collapse multiplier is $\rho^A=1$ *already for* $\widetilde{A}$, with no renormalisation. The proof is one line of symmetry — s_{ij} *permutes* the argument multiset $w_{\mathbf{M}}$ instead of deleting from it — and that single structural difference is the whole explanation of why the pair divisors carried poles and these do not: the pole was the mismatch $\rho^2-\rho$, and $1^2-1=0$.

Theorem A (both types B and C). $\mathcal{L}_{ij}:=\bigoplus_{\mathbf{I}}\mathcal{O}_{ij}\phi_{\mathbf{I}}$ is *equal* to the A-type GKM lattice $\mathcal{G}_{ij}=\{f:f(\mathbf{M})\equiv f(s_{ij}\mathbf{M})\}$. Both have colength exactly 2^{N-2} in $\mathcal{O}_{ij}^{2^{[N]}}$ — on the GKM side one condition per 2-element s_{ij}-orbit, on the span side $\sum_{\mathbf{I}}\operatorname{ord}\widetilde{A}_{\mathbf{I},\mathbf{I}}=\#\{\mathbf{I}:i\notin\mathbf{I},\,j\in\mathbf{I}\}$, and these are the same 2^{N-2}.

Theorem B. Since $\mathcal{G}_{ij}$ is visibly a ring, every structure constant is regular on every A-type divisor — for $\widetilde{A}$, and for $\Psi\widetilde{A}$ for *every* $\Psi\in\mathcal{G}_{ij}^{\times}$. Lesioning the hypothesis (V7) moved it off the point-product family in both directions: a *non*-product $\Psi=1+(w_i-w_j)$ on one point survives (it is $\equiv1$, hence in $\mathcal{G}^{\times}$ anyway), while a Ψ that is a unit at every point but $\not\equiv1$ produces six A-type poles. So the hypothesis is $\mathcal{G}^{\times}$-membership, not point-indexedness and not pointwise invertibility. In particular the A-type direction imposes *no condition on* χ within the point-product family $\Psi_\chi(\mathbf{M})=\prod_{m\in\mathbf{M}}\chi(w_m)$, which is the exact complement of Theorem C of `2026-10-17-c2`: (COB) and (SR) are the only conditions there are.

The pre-registered branch fired, against my stated expectation. The brief pre-registered: *if $\widetilde{A}$ has no A-type poles then gap 2 was never a gap; report that plainly and do not dress a vacuity as a result.* $\widetilde{A}$ has none — the baseline census's 13/115 non-Laurent constants are **pair-divisor poles only**, zero A-type, and all 35 nonzero basis entries at $N \le 3$ are Laurent. So there was never an A-type defect for Ψ to create or destroy, and gap 2 *as a difference question* is vacuous. What is not vacuous is the regularity itself, which was not previously proved, and its mechanism. I expected the opposite and say so in §5; this is the third time this week I have mispredicted a normaliser's reach.

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1 Setup

I keep the notation of `2026-10-17-c2-coboundary-and-the-removable-defect.tex` (commit `7002d00`), including the corrected divisor nomenclature of its Remark 2: the divisors are named by the *root* that cuts them, not by a root length, because long/short is a function of h and these statements quantify over h.

Let $F := \mathbb{C}(q)(w_1, \dots, w_N)$ and $P := 2^{[N]}$, the set of subsets of $[N] = \{1, \dots, N\}$; $|P| = 2^N$. Under the identification of P with the T-fixed points of the Lagrangian Grassmannian $LG(N, 2N)$, we regard a basis element $\phi_{\mathbf{I}}$ as the function

$$\phi_{\mathbf{I}} : P \to F, \qquad \phi_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}_{\mathbf{M},\mathbf{I}},$$

where $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \widetilde{W}^{C}_{N,|\mathbf{M}|,|\mathbf{I}|}(w_{\mathbf{M}} \mid w \mid \mathbf{I})$ is Motegi's $\mathcal{E}^C$-normalised type-C weight function evaluated at the specialisation $u = w_{\mathbf{M}} := (w_m)_{m \in \mathbf{M}}$, via the stability relation (`2610.15344`, eq. `typeC-stability-s-ge-r`, L2822–2828). Let $\mathcal{A} := F^P$ with *pointwise* addition and multiplication; this is a commutative F-algebra with unit the constant function 1.

(M3) Triangularity. $\widetilde{A}_{\mathbf{M},\mathbf{I}} \neq 0$ only if $\mathbf{M} \succeq \mathbf{I}$ in Motegi's order, defined at **L582–589** as $|\mathbf{M}| \ge |\mathbf{I}|$ and $m_{s-r+j} \ge i_j$ for all j; the vanishing itself is proved at L670–675. (Locator corrected in this session: my engine's docstring, and a first draft of this paper, cited "L640–650" for the order. L640–650 lies *inside* the vanishing proof and contains no definition. Fixed in all four copies of the engine, not only the one in use.) Also $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ when $|\mathbf{M}| < |\mathbf{I}|$ (L2843).

(M4) The diagonal. $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ is the product over the positive roots of C_N: $\widetilde{A}_{\mathbf{I},\mathbf{I}} = \prod_{\alpha\in\mathrm{Inv}(\lambda)}(1-e^{-\alpha})\prod_{\alpha\in T(\lambda)}(1-qe^{+\alpha})$ (eq. `normalizationWC`, L1078), with $\mathrm{Inv}(\lambda)=\{\alpha\in\Delta^+ : s_\alpha\lambda<\lambda\}$ and $T(\lambda)=\{\alpha\in\Delta^+ : s_\alpha\lambda>\lambda\}$ written out explicitly at L997–1010 and L1017–1030:

$$\widetilde{A}_{\mathbf{I},\mathbf{I}} = \prod_{1\le a\le b\le r}\Big(1-\tfrac{1}{w_{i_a}w_{i_b}}\Big)\cdot\prod_{\substack{c\in\mathbf{I}^c\\ i_l>c}}\Big(1-\tfrac{w_c}{w_{i_l}}\Big)\cdot\prod_{\substack{c\le d\\ c,d\in\mathbf{I}^c}}(1-qw_cw_d)\cdot\prod_{\substack{c\in\mathbf{I}^c\\ i_l<c}}\Big(1-q\tfrac{w_c}{w_{i_l}}\Big). \tag{1}$$

By (M3) and (M4) the matrix $\Phi := (\widetilde{A}_{\mathbf{M},\mathbf{I}})_{\mathbf{M},\mathbf{I}\in P}$ is triangular with respect to any linear extension of $\succeq$, with nonzero diagonal, so $\det\Phi = \pm\prod_{\mathbf{I}}\widetilde{A}_{\mathbf{I},\mathbf{I}}\neq 0$. Hence $\{\phi_{\mathbf{I}}\}_{\mathbf{I}\in P}$ is an F-basis of $\mathcal{A}=F^P$, and the *structure constants* are the $c^{\nu}_{\lambda\mu}\in F$ with $\phi_\lambda\phi_\mu=\sum_\nu c^{\nu}_{\lambda\mu}\phi_\nu$.

Definition 1 (the A-type divisor and its local ring)**.** Fix $i<j$. In $R:=\mathbb{C}(q)[w_1^{\pm},\dots,w_N^{\pm}]$ the element w_i-w_j is irreducible, so $\mathfrak{p}_{ij}:=(w_i-w_j)$ is a height-one prime and

$$\mathcal{O}_{ij} := R_{\mathfrak{p}_{ij}} \subset F$$

is a discrete valuation ring with uniformiser $t:=w_i-w_j$, valuation $\mathrm{ord}=\mathrm{ord}_{(w_i-w_j)}$, residue field k_{ij} and fraction field F. Call $\{w_i=w_j\}$ the ***A*-type divisor**; it is cut out by the character w_i/w_j, i.e. by the root e_i-e_j.

Definition 2 (the reflection, and which points it moves)**.** Let s_{ij} act on P by exchanging membership of i and j:

$$s_{ij}\mathbf{M} := \begin{cases}\mathbf{M}\triangle\{i,j\} & |\mathbf{M}\cap\{i,j\}|=1,\\ \mathbf{M} & \text{otherwise.}\end{cases}$$

Write $P^{\mathrm{mv}}:=\{\mathbf{M} : |\mathbf{M}\cap\{i,j\}|=1\}$. Then $|P^{\mathrm{mv}}|=2^{N-1}$ and s_{ij} partitions it into exactly 2^{N-2} orbits of size 2; the remaining 2^{N-1} points are fixed.

Remark 3 (the fixed-point/curve model, cross-checked against an independent count)*.* s_{ij} is the reflection in the root e_i-e_j, acting on the T-fixed points of $LG(N,2N)$; the associated T-curve has weight w_i/w_j, so "$\equiv$ along that curve" means congruence modulo $1-w_i/w_j$, a unit multiple of t. The same model assigns to e_a+e_b the map $\mathbf{M}\mapsto\mathbf{M}\setminus\{a,b\}$ (when $\{a,b\}\subseteq\mathbf{M}$) and to $2e_m$ the map $\mathbf{M}\mapsto\mathbf{M}\setminus\{m\}$ — which is precisely the shape of (QC) and (SRC) in `2026-10-17-c2`, so the model is consistent with what is already proved. Independent check of the *count*: at $N=2$ the model predicts $1+1+2+2=6$ curves over the four reflections of $W(C_2)$, and Rick's $LG(2,4)$ GKM checker (UID 824, a different author's instrument) tests exactly six congruences.

2 Lemma A1: the A-type multiplier is 1, by symmetry

Lemma 4 (u-symmetry)**.** *For all N,n,r the normalised weight function $\widetilde{W}^{X}_{N,n,r}(u \mid w \mid \mathbf{I})$ is a symmetric function of $u_1,\dots,u_n$, for $X=B$ and $X=C$.*

Proof. Motegi's Motherfunction (eq. `Motherfunction`, L474–487) is

$$F_{N,n,r}(u\mid w\mid\mathbf{I}) = \frac{1}{(n-r)!}\Big(\prod_{k=1}^{n}u_k^{-(N+n)}\Big)\sum_{\sigma\in S_n}T\big(u_{\sigma(1)},\dots,u_{\sigma(n)}\big),$$

for a fixed expression T depending on $w,\mathbf{I},\beta,\gamma$. A sum of $T\circ\sigma$ over the *full* symmetric group is symmetric whatever T is, and the prefactor $\prod_k u_k^{-(N+n)}$ is symmetric; hence F is symmetric

in u. The type-C normaliser $\mathcal{E}^C$ (eq. `typeCnormalizationfactor`, L1057–1062) is built from $\prod_{j,k}(1-qu_k/u_j)$, $\prod_{j,k}(1-qu_jw_k)$, $\prod_{j<k}(1-qu_ju_k)$, a u-free factor and $(1-q)^{n-r}$ — each symmetric in u. For type B, W^B carries the extra symmetric prefactor $\prod_k(1+1/u_k)^{-1}$ and $\beta,\gamma=1,-1$, and $\mathcal{E}^B$ (L1361–1367) adds $\prod_j(1-qu_j)$ and $\prod_j(1-qu_j^2)$, again symmetric. Therefore $\widetilde{W}^X = W^X/\mathcal{E}^X$ is symmetric in u. □

Lemma 5 (regularity on A-type divisors)**.** *$\widetilde{A}_{\mathbf{M},\mathbf{I}} \in \mathcal{O}_{ij}$ for every $\mathbf{M},\mathbf{I} \in P$ and every $i<j$; the same holds for type B.*

Proof. Two sources of a possible pole must be excluded.

(a) The normaliser. $\widetilde{A} = W^X \cdot \mathcal{E}^{X\,-1}$, so poles can come only from zeros of the numerator of $\mathcal{E}^X$, namely from $\prod_{j,k}(1-qu_k/u_j)$, $\prod_{j,k}(1-qu_jw_k)$ and (type B) $\prod_j(1-qu_j)$. At $u=w_{\mathbf{M}}$ these vanish on $\{w_b=qw_a\}$, $\{qw_aw_b=1\}$ and $\{qw_a=1\}$ respectively. Each carries a q; none of them contains the generic point of $\{w_i=w_j\}$, since q is transcendental over $\mathbb{C}(w)$. So $\operatorname{ord}(\mathcal{E}^X)=0$ and the normaliser contributes nothing.

(b) The Motherfunction. In $T(v_1,\dots,v_n)$ the only factors with a denominator are

$$\frac{qv_a-v_b}{v_a-v_b}\quad (a<b\le r)\qquad\text{and}\qquad \frac{qv_b-v_a}{v_b-v_a}\quad (a\le r<b),$$

one for each slot pair $\{a,b\}$ with $\min(a,b)\le r$; each is a simple pole on $\{v_a=v_b\}$. Fix two distinct elements $p\neq p'$ of $\mathbf{M}$, so $u_p=w_p$ and $u_{p'}=w_{p'}$ are distinct variables. For $\sigma\in S_n$ let $a:=\sigma^{-1}(p)$, $b:=\sigma^{-1}(p')$ be the slots carrying them, and put

$$\sigma' := \sigma\circ(a\,b).$$

Since σ is a bijection, $a\neq b$ are well defined, so $\sigma\mapsto\sigma'$ is a fixed-point-free involution of S_n. I claim

$$\operatorname{Res}_{w_p=w_{p'}}\Big[\,T(u_\sigma)+T(u_{\sigma'})\,\Big]=0.$$

Write $x:=u_{\sigma(a)}$, $y:=u_{\sigma(b)}$, so $\{x,y\}=\{w_p,w_{p'}\}$. The two terms differ *only* by exchanging the values in slots a and b. Every factor other than the one attached to the pair $\{a,b\}$ is regular at $x=y$: the pair $\{a,b\}$ is the unique slot pair whose two values are $w_p,w_{p'}$, because σ is injective. Moreover at $x=y$ slots a and b carry equal values, so each of those other factors takes *literally the same value* in $T(u_\sigma)$ and in $T(u_{\sigma'})$; call that common value C. For the dangerous factor, if $a<b\le r$ then

$$\operatorname{Res}_{x=y}\frac{qx-y}{x-y}=x(q-1),\qquad \operatorname{Res}_{x=y}\frac{qy-x}{y-x}=-\,y(q-1)=-\,x(q-1),$$

and if $a\le r<b$ the same two values occur with the roles exchanged. Either way the two residues are $C\cdot x(q-1)$ and $-C\cdot x(q-1)$, and they cancel. Summing over the $n!/2$ involution pairs, $\sum_\sigma T(u_\sigma)$ has no pole on $\{w_p=w_{p'}\}$. If $\min(a,b)>r$ the pair carries no such factor and both terms are already regular.

Taking $\{p,p'\}=\{i,j\}$ when $\{i,j\}\subseteq\mathbf{M}$ settles the only case in which w_i-w_j can occur as a difference of two u-arguments at all. If $|\mathbf{M}\cap\{i,j\}|\le 1$ then w_i-w_j is never of the form u_a-u_b, and (b) is vacuous. Together with (a), $\operatorname{ord}(\widetilde{A}_{\mathbf{M},\mathbf{I}})\ge 0$. □

Lemma 6 (A1: the A-type GKM congruence, with multiplier exactly 1)**.** *For all N, all $i<j$, all $\mathbf{M}\in P$ and all $\mathbf{I}\in P$,*

$$\widetilde{A}_{\mathbf{M},\mathbf{I}} \;\equiv\; \widetilde{A}_{s_{ij}\mathbf{M},\mathbf{I}}\pmod{w_i-w_j},$$

for type C and for type B. Equivalently $\rho^A=1$: no renormalisation is needed, and none can change this.

Proof. If $|\mathbf{M}\cap\{i,j\}|\neq 1$ then $s_{ij}\mathbf{M}=\mathbf{M}$ and there is nothing to prove. So assume without loss of generality $i\in\mathbf{M}$, $j\notin\mathbf{M}$, and put $\mathbf{M}^\circ:=\mathbf{M}\setminus\{i\}$, so $s_{ij}\mathbf{M}=\mathbf{M}^\circ\sqcup\{j\}$. In particular $|s_{ij}\mathbf{M}|=|\mathbf{M}|$: the reflection **preserves the cardinality** and therefore the integers $n=|\mathbf{M}|$ and $r=|\mathbf{I}|$ are the same on both sides.

By Lemma 5 both sides lie in $\mathcal{O}_{ij}$, so both may be reduced modulo $t=w_i-w_j$, i.e. evaluated under the substitution $\varepsilon: w_j\mapsto w_i$. Write $G(u_1,\dots,u_n):=\widetilde{W}^X_{N,n,r}(u\mid w\mid \mathbf{I})$, a function of n arguments with the alphabet w and the label $\mathbf{I}$ as parameters. Then

$$\widetilde{A}_{\mathbf{M},\mathbf{I}}=G\big(w_i,(w_m)_{m\in\mathbf{M}^\circ}\big),\qquad \widetilde{A}_{s_{ij}\mathbf{M},\mathbf{I}}=G\big(w_j,(w_m)_{m\in\mathbf{M}^\circ}\big),$$

where by Lemma 4 the order of the arguments is immaterial. Apply ε to both. It acts on the parameters w identically in the two cases, and on the argument tuples it gives

$$\varepsilon\big(w_i,(w_m)_{m\in\mathbf{M}^\circ}\big)=\big(w_i,(\varepsilon w_m)_{m\in\mathbf{M}^\circ}\big)=\varepsilon\big(w_j,(w_m)_{m\in\mathbf{M}^\circ}\big),$$

the middle equality because $j\notin\mathbf{M}^\circ$, so ε does not touch the tail, and $\varepsilon(w_j)=w_i$. The two argument multisets therefore become *equal*, and the two sides become the same expression. □

Remark 7 (why the pair divisors behaved differently — the whole mechanism). Lemma 6 is short, and the reason it is short is the content. Compare the three root families by what they do to $\mathbf{M}$:

root	action on $\mathbf{M}$	effect on $\lvert\mathbf{M}\rvert$
e_i-e_j	$\mathbf{M}\mapsto\mathbf{M}\triangle\{i,j\}$	**preserved**
e_a+e_b	$\mathbf{M}\mapsto\mathbf{M}\setminus\{a,b\}$	drops by 2
$h\cdot e_m$	$\mathbf{M}\mapsto\mathbf{M}\setminus\{m\}$	drops by 1

Only in the A-type case is the collision a *permutation of the argument multiset* $w_{\mathbf{M}}$; u-symmetry then supplies the congruence with multiplier 1, for free and for every h. In the other two cases the number of arguments changes, symmetry says nothing, and the multiplier has to be *computed* from a genuine stability/collapse relation.

What this does and does not explain. It explains why the A-type case cost nothing: it is the one family whose multiplier is forced without computing it. It does *not* predict which families carry poles, and I want to be precise about that, because the stronger reading is tempting and false. Changing the arity is *necessary but not sufficient* for a nontrivial multiplier: the boundary root $h\cdot e_m$ also drops $|\mathbf{M}|$, and its multiplier, once computed, is **exactly** 1 (Theorem (SRC) of `2026-10-17-c2`). So the boundary family is the counterexample to "arity changes $\Rightarrow\rho\neq 1$". The honest statement is: arity preserved $\Rightarrow\rho=1$ by symmetry; arity changed $\Rightarrow$ no information, go and compute, and the two computations landed differently ($\rho^X\neq 1$ for e_a+e_b, $\rho=1$ for $h\cdot e_m$).

Given the multipliers, though, the pole census follows exactly. The product of two basis elements collides with multiplier ρ^2 while their span collides with ρ, so the obstruction is $\rho^2-\rho=\rho(\rho-1)$, which vanishes precisely when $\rho\in\{0,1\}$. On the pair divisors $\rho=g\neq 1$ and there are poles (13/115 at $N=3$); on the boundary divisors $\rho=1$ and there are none; on the A-type divisors $\rho^A=1$ and $1^2-1=0$. *The poles were never about the A-direction, and the renormalisation ψ was never needed there.*

3 Theorem A: the span is exactly the GKM lattice

Fix $i<j$ and abbreviate $\mathcal{O}:=\mathcal{O}_{ij}$, $s:=s_{ij}$, $t:=w_i-w_j$, $k:=k_{ij}$.

Definition 8.

$$\mathcal{G} := \big\{ f \in \mathcal{O}^P \ : \ f(\mathbf{M}) \equiv f(s\mathbf{M}) \pmod{t} \ \ \forall\, \mathbf{M} \in P \big\}, \qquad \mathcal{L} := \bigoplus_{\mathbf{I} \in P} \mathcal{O}\, \phi_{\mathbf{I}}.$$

Lemma 9 ($\mathcal{G}$ is a ring)**.** *$\mathcal{G}$ is an $\mathcal{O}$-subalgebra of $\mathcal{O}^P$ containing* 1*.*

Proof. The constant function 1 satisfies the congruences. $\mathcal{G}$ is an $\mathcal{O}$-submodule because the conditions are $\mathcal{O}$-linear. For closure under multiplication, for $f, g \in \mathcal{G}$ and any $\mathbf{M}$,

$$(fg)(\mathbf{M}) - (fg)(s\mathbf{M}) = f(\mathbf{M})\big[g(\mathbf{M}) - g(s\mathbf{M})\big] + \big[f(\mathbf{M}) - f(s\mathbf{M})\big] g(s\mathbf{M}) \in t\mathcal{O},$$

since all four of $f(\mathbf{M}), g(s\mathbf{M})$ and the two brackets lie in $\mathcal{O}$ and the brackets lie in $t\mathcal{O}$. □

Lemma 10 (colength of $\mathcal{G}$)**.** $\operatorname{length}_{\mathcal{O}}\big(\mathcal{O}^P/\mathcal{G}\big) = 2^{N-2}$.

Proof. Choose one representative $\mathbf{M}_\alpha$ from each of the 2^{N-2} two-element s-orbits α and consider

$$\delta : \mathcal{O}^P \longrightarrow k^{\,2^{N-2}}, \qquad \delta(f)_\alpha := \big[f(\mathbf{M}_\alpha) - f(s\mathbf{M}_\alpha) \big] \bmod t.$$

This is $\mathcal{O}$-linear, and $\ker \delta = \mathcal{G}$: the congruence at $s\mathbf{M}_\alpha$ is the negative of the one at $\mathbf{M}_\alpha$, and at an s-fixed point it is $0 \equiv 0$. It is surjective: for a target vector $(\bar{c}_\alpha)$, lift each $\bar{c}_\alpha$ to $c_\alpha \in \mathcal{O}$ and take $f = \sum_\alpha c_\alpha \mathbf{1}_{\mathbf{M}_\alpha}$, whose δ is $(\bar{c}_\alpha)$ because the points $\mathbf{M}_\alpha, s\mathbf{M}_\alpha$ are pairwise distinct across orbits. Hence $\mathcal{O}^P/\mathcal{G} \cong k^{2^{N-2}}$, of length 2^{N-2}. □

Lemma 11 (colength of $\mathcal{L}$)**.** $\operatorname{length}_{\mathcal{O}}\big(\mathcal{O}^P/\mathcal{L}\big) = \sum_{\mathbf{I} \in P} \operatorname{ord} \widetilde{A}_{\mathbf{I},\mathbf{I}} = 2^{N-2}$.

Proof. By Lemma 5 the matrix $\Phi = (\widetilde{A}_{\mathbf{M},\mathbf{I}})$ has entries in $\mathcal{O}$, and $\mathcal{L}$ is exactly its column span, i.e. the image of $\Phi : \mathcal{O}^P \to \mathcal{O}^P$. Since $\det \Phi \neq 0$, the map is injective, and over a discrete valuation ring the colength of the image of an injective endomorphism of a free module of finite rank is the valuation of its determinant (Smith normal form). By (M3)+(M4), $\det \Phi = \pm \prod_{\mathbf{I}} \widetilde{A}_{\mathbf{I},\mathbf{I}}$, so $\operatorname{length}(\mathcal{O}^P/\mathcal{L}) = \sum_{\mathbf{I}} \operatorname{ord} \widetilde{A}_{\mathbf{I},\mathbf{I}}$.

Now evaluate $\operatorname{ord} \widetilde{A}_{\mathbf{I},\mathbf{I}}$ from (1), factor by factor, by asking which factors contain the generic point of $\{w_i = w_j\}$. The factors $\big(1 - q w_c w_d\big)$ and $\big(1 - q w_c / w_{i_l}\big)$ carry a q and so do not (same argument as Lemma 5(a)). The factor $\big(1 - 1/(w_{i_a} w_{i_b})\big)$ cuts the pair divisor $\{w_{i_a} w_{i_b} = 1\}$ (or, at $a = b$, the boundary divisor $\{w_{i_a}^2 = 1\}$), an irreducible divisor distinct from $\{w_i = w_j\}$, so it does not either. The only remaining factors are the A-type ones

$$\Big(1 - \frac{w_c}{w_{i_l}}\Big), \qquad c \in \mathbf{I}^c,\ i_l \in \mathbf{I},\ i_l > c,$$

and such a factor vanishes on $\{w_i = w_j\}$ if and only if $\{c, i_l\} = \{i, j\}$. Given $i < j$ and the constraint $i_l > c$, this forces $c = i$ and $i_l = j$. Each such factor is simple and occurs at most once. Therefore

$$\operatorname{ord} \widetilde{A}_{\mathbf{I},\mathbf{I}} = \begin{cases} 1, & i \notin \mathbf{I} \text{ and } j \in \mathbf{I}, \\ 0, & \text{otherwise}, \end{cases} \qquad \text{so} \qquad \sum_{\mathbf{I} \in P} \operatorname{ord} \widetilde{A}_{\mathbf{I},\mathbf{I}} = \#\{\mathbf{I} : i \notin \mathbf{I},\ j \in \mathbf{I}\} = 2^{N-2}. \qquad \square$$

Lemma 12 (the A-type blocks of the diagonal do not know the type)**.** *Lemma 11 holds verbatim for type B: for all $\mathbf{I}$,* $\operatorname{ord}_{(w_i - w_j)} \widetilde{A}^B_{\mathbf{I},\mathbf{I}} = [\, i \notin \mathbf{I} \text{ and } j \in \mathbf{I}\,]$ *and* $\sum_{\mathbf{I}} \operatorname{ord} \widetilde{A}^B_{\mathbf{I},\mathbf{I}} = 2^{N-2}$.

Proof. Motegi gives the same root-product shape in both types — $\prod_{\mathrm{Inv}(\lambda)}(1-e^{-\alpha})\prod_{T(\lambda)}(1-qe^{+\alpha})$, eq. `normalizationWC` (L1078) and eq. `normalizationWB` (L1386) — and writes out Inv and T explicitly in each case: type C at L997–1010 and L1017–1030, type B at L1307–1320 and L1322–1340. Read first-hand this session, the two decompositions are

$$\begin{aligned}
\mathrm{Inv}(\lambda) \text{ type } C &= \{2t_k\}_{k\in\mathbf{I}} \;\cup\; \{t_j{+}t_k\}_{j<k;\, j,k\in\mathbf{I}} \;\cup\; \{\mathbf{t_k{-}t_j}\}_{\mathbf{j<k};\ j\notin\mathbf{I},\,k\in\mathbf{I}},\\
\mathrm{Inv}(\lambda) \text{ type } B &= \;\{t_k\}_{k\in\mathbf{I}} \;\cup\; \{t_j{+}t_k\}_{j<k;\, j,k\in\mathbf{I}} \;\cup\; \{\mathbf{t_k{-}t_j}\}_{\mathbf{j<k};\ j\notin\mathbf{I},\,k\in\mathbf{I}},\\
T(\lambda) \text{ type } C &= \{2t_k\}_{k\notin\mathbf{I}} \;\cup\; \{t_j{+}t_k\}_{j<k;\, j,k\notin\mathbf{I}} \;\cup\; \{\mathbf{t_k{-}t_j}\}_{\mathbf{j<k};\ j\in\mathbf{I},\,k\notin\mathbf{I}},\\
T(\lambda) \text{ type } B &= \;\{t_k\}_{k\notin\mathbf{I}} \;\cup\; \{t_j{+}t_k\}_{j<k;\, j,k\notin\mathbf{I}} \;\cup\; \{\mathbf{t_k{-}t_j}\}_{\mathbf{j<k};\ j\in\mathbf{I},\,k\notin\mathbf{I}}.
\end{aligned}$$

The A-type members (bold) are *the same subsets of the same root lattice*, and the corresponding product blocks are correspondingly identical in the two papers' own notation:

$$\prod_{j\in\mathbf{I}^c}\prod_{l:\,i_l>j}\left(1-w_{i_l}^{-1}w_j\right) \quad \text{from Inv,} \qquad \prod_{j\in\mathbf{I}^c}\prod_{l:\,i_l<j}\left(1-q\,w_{i_l}^{-1}w_j\right) \quad \text{from } T.$$

The *entire* type-dependence sits on the boundary root: $2t_k$ versus t_k, i.e. $(1-w_{i_j}^{-2})$ versus $(1-w_{i_j}^{-1})$ in Inv and $(1-qw_k^2)$ versus $(1-qw_k)$ in T — which is also why type C can absorb its $j=k$ term into $\prod_{j\le k}$ while type B carries a separate $\prod_j(1-w_{i_j}^{-1})$ and a strict $\prod_{j<k}$. Neither $\{w_{i_j}=1\}$ nor $\{qw_k=1\}$ contains the generic point of $\{w_i=w_j\}$, so the factor-by-factor count in the proof of Lemma 11 is unchanged, and only the bold blocks contribute. □

Remark 13 (the type enters only through h, for the third time). Lemma 12 is the same phenomenon as the previous paper's thesis and as this one's Remark 7: type B and type C differ *only* in the boundary root $h\cdot e_m$, and every statement about the A-type direction is therefore type-free. It is worth recording that I did *not* obtain this by assuming it. I was about to file the type-B arm as a gap resting on a measured colength, and went to check a *different* locator — the one for Motegi's order — which turned out to be wrong; finding the right one put me in the neighbourhood of `normalizationWB`, which I then read. Reusing (1) for type B *without* reading L1307–1340 would have been the exact shape of the $\mathcal{E}^B$ error of 2026-10-17 (a type-C formula applied to type B, green and wrong); the difference between that error and this lemma is four lines of reading, not an inference.

Theorem 14 (the span *is* the A-type GKM lattice)**.** *For every $N\ge 2$, every $i<j$, and* both *types B and C,*

$$\mathcal{L}_{ij} \;=\; \mathcal{G}_{ij}.$$

Proof. $\mathcal{L}\subseteq\mathcal{G}$: each $\phi_{\mathbf{I}}$ lies in $\mathcal{O}^P$ by Lemma 5 and satisfies the congruences by Lemma 6, and $\mathcal{G}$ is an $\mathcal{O}$-module. Both have colength 2^{N-2} in $\mathcal{O}^P$ by Lemmas 10 and 11 (type C), resp. Lemma 12 (type B), so

$$\mathrm{length}(\mathcal{G}/\mathcal{L}) = \mathrm{length}(\mathcal{O}^P/\mathcal{L}) - \mathrm{length}(\mathcal{O}^P/\mathcal{G}) = 2^{N-2}-2^{N-2}=0,$$

the subtraction being legitimate because both lengths are finite. A module of length 0 is zero, so $\mathcal{L}=\mathcal{G}$. □

Remark 15 (the two 2^{N-2}'s are not the same count wearing one name). On the GKM side 2^{N-2} counts *two-element orbits of s_{ij} on P*. On the span side it counts *labels* $\mathbf{I}$ *with* $i\notin\mathbf{I}$, $j\in\mathbf{I}$. These index sets are genuinely different — an orbit is an unordered pair of points, a label is a single subset — and the bijection between them ($\{\mathbf{M}, s\mathbf{M}\}\mapsto$ the member containing j, intersected with the complement of $\{i\}$) is not used anywhere in the proof. The two numbers are computed by different routes and then compared, which is the only way the comparison carries information.

4 Theorem B: no condition on χ

Definition 16. For $\chi \in F(v)$ put $\Psi_\chi(\mathbf{M}) := \prod_{m\in\mathbf{M}} \chi(w_m)$ and $\widehat{A}^{\chi}_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}}\,\Psi_\chi(\mathbf{M})$, i.e. $\hat{\phi}^{\chi}_{\mathbf{I}} = \Psi_\chi \cdot \phi_{\mathbf{I}}$ as functions on P. Call χ *ij*-**admissible** if $\chi(w_m) \in \mathcal{O}^{\times}_{ij}$ for every $m \in [N]$, i.e. $\chi(w_m)$ is regular and non-vanishing at the generic point of $\{w_i = w_j\}$.

Lemma 17 (Ψ_χ is a unit of $\mathcal{G}$ — and this is why it carries no information)**.** *If χ is ij-admissible then $\Psi_\chi \in \mathcal{G}^{\times}$, i.e. both Ψ_χ and Ψ_χ^{-1} lie in $\mathcal{G}$.*

Proof. $\Psi_\chi(\mathbf{M})$ and $\Psi_\chi(\mathbf{M})^{-1}$ are finite products of the units $\chi(w_m)^{\pm 1}$, hence lie in $\mathcal{O}^{\times} \subset \mathcal{O}$. For the congruence take $\mathbf{M}$ with $i \in \mathbf{M}$, $j \notin \mathbf{M}$ and $\mathbf{M}^{\circ} = \mathbf{M} \setminus \{i\}$; then

$$\Psi_\chi(\mathbf{M}) - \Psi_\chi(s\mathbf{M}) = \Psi_\chi(\mathbf{M}^{\circ})\big[\chi(w_i) - \chi(w_j)\big] \in t\mathcal{O},$$

because $\chi(w_i) - \chi(w_j) \in t\mathcal{O}$: writing $\chi = \chi_1/\chi_2$ with $\chi_2(w_i), \chi_2(w_j)$ units, the difference is $\big(\chi_1(w_i)\chi_2(w_j) - \chi_1(w_j)\chi_2(w_i)\big)/\chi_2(w_i)\chi_2(w_j)$ and the numerator is a polynomial expression vanishing under $w_j \mapsto w_i$. The identical computation with χ^{-1} in place of χ gives $\Psi_\chi^{-1} \in \mathcal{G}$. □

Remark 18 (Task 1 of the brief, and its exact epistemic status)*.* Lemma 17 is the observation the brief flagged at Task 1 and instructed me to grade honestly, so I do. **It carries zero bits about ψ.** The proof uses nothing about χ beyond admissibility; it goes through verbatim for $\chi(v) = v$, for $\chi(v) = 1$, for $\chi(v) = \prod$(anything regular and nonvanishing). The pre-registered "non-invariant" control $\prod_{m\in\mathbf{M}} w_m + |\mathbf{M}|$ failed to fire at the 2026-10-18 wake for exactly this reason, and that silent control was the finding: the invariance is a property of the *shape* $\prod_{m\in\mathbf{M}} f(w_m)$, not of ψ. Measured there: 0 violations in 96 instances at $N = 4$, $h = 2$, with the off-divisor arm firing 48/48.

And yet it is logically indispensable here, which is the part worth recording. Having zero discriminating power between $\widehat{A}$ and $\widetilde{A}$ is precisely what makes Theorem 19 hold for the *whole* family rather than for ψ alone. A fact can be useless as evidence and load-bearing as a hypothesis; the error would be to let the second tempt me into reporting the first as support for ψ. It is not. Cf. the one-line version in §5.

Theorem 19 (every structure constant is regular on every A-type divisor)**.** *Fix $i < j$ and let $\Psi \in \mathcal{G}^{\times}_{ij}$ be* any *invertible element of the ring $\mathcal{G}_{ij}$ — not necessarily a point-indexed product. Let $\hat{\phi}_{\mathbf{I}} := \Psi\,\phi_{\mathbf{I}}$ and let $\hat{c}^{\nu}_{\lambda\mu}$ be the resulting structure constants. Then for all λ, μ, ν,*

$$\hat{c}^{\nu}_{\lambda\mu} \in \mathcal{O}_{ij}.$$

Proof. Let $\hat{\mathcal{L}} := \bigoplus_{\mathbf{I}} \mathcal{O}\hat{\phi}_{\mathbf{I}} = \Psi \cdot \mathcal{L}$. By Theorem 14 and because Ψ is a unit of the ring $\mathcal{G}$ (Lemma 9),

$$\hat{\mathcal{L}} = \Psi\,\mathcal{L} = \Psi\,\mathcal{G} = \mathcal{G},$$

so $\hat{\mathcal{L}}$ is a ring. Hence $\hat{\phi}_\lambda\hat{\phi}_\mu \in \hat{\mathcal{L}}$, and expanding it in the $\mathcal{O}$-basis $\{\hat{\phi}_\nu\}$ of $\hat{\mathcal{L}}$ gives coefficients in $\mathcal{O}$. These coefficients are the $\hat{c}^{\nu}_{\lambda\mu}$, by uniqueness of the expansion in the F-basis $\{\hat{\phi}_\nu\}$ of F^P. □

Corollary 20 (the point-product family)**.** *If χ is ij-admissible for all $i < j$, then every structure constant of $\Psi_\chi\widetilde{A}$ is regular on every A-type divisor. In particular this holds for $\widetilde{A}$ itself ($\chi = 1$) and for $\widehat{A} = \Psi_\psi\widetilde{A}$, and the A-type behaviour is the same for every member of the family.*

Proof. Lemma 17 puts Ψ_χ in $\mathcal{G}^{\times}_{ij}$ for each $i < j$; apply Theorem 19 for each. □

Remark 21 (the hypothesis is $\mathcal{G}^\times$-membership, and *not* either of the two things it is easy to mistake it for). I stated Theorem 19 for the point-product family first and then lesioned it (V7), which moved the hypothesis twice.

- It is **not** "Ψ is a point-indexed product". The lesion $\Psi = 1 + (w_1 - w_2)$ on the single point $\mathbf{M} = \{1\}$ and 1 elsewhere is *not* of the form $\prod_{m\in\mathbf{M}} \chi(w_m)$, and the conclusion holds anyway (A-type set empty) — because $1+(w_1-w_2) \equiv 1$, so that Ψ lies in $\mathcal{G}^\times$ regardless. A surviving lesion is a stronger theorem for free, and this is why Theorem 19 is now stated for $\mathcal{G}^\times$ with the product family demoted to Corollary 20.
- It is **not** "Ψ is a pointwise unit" either. The lesion $\Psi = 2$ on $\mathbf{M} = \{1\}$ and 1 elsewhere is a unit at every point, with a unit inverse, and it produces **six** A-type poles — because $2 \not\equiv 1$, so it fails the congruence and is not in $\mathcal{G}$ at all.

The two lesions differ only in whether the perturbation is $\equiv 1$ modulo the uniformiser, and they land on opposite sides. That pair is the sharp control for this hypothesis: together they show $\mathcal{G}^\times$ is neither too strong (the first survives) nor too weak (the second breaks).

Corollary 22 (the renormalisation-invariance statement)**.** *The lattice $\mathcal{L}_{ij}$ is independent of χ within the admissible family: $\bigoplus_{\mathbf{I}} \mathcal{O}_{ij}\, \Psi_\chi \phi_{\mathbf{I}} = \mathcal{G}_{ij}$ for every ij-admissible χ. Consequently the A-type divisors impose **no condition** on χ.*

Remark 23 (the exact complement of Theorem C of `2026-10-17-c2`). Theorem C of the previous paper determined the admissible renormalisations by *two* conditions and showed both are necessary: (COB) $\chi(x)\chi(1/x) = \rho^{-1}|_H$ for the pair divisors, and (SR) $\chi(\varepsilon) = 1$ for $\varepsilon^h = 1$ for the boundary divisors. It did not know whether a *third* condition was waiting at the A-type divisors. Corollary 22 says there is none: (COB) and (SR) are the complete list. This sharpens Theorem C from "these two conditions suffice for the two families examined" to "these two conditions are the whole story across all three families of positive roots", and it is the statement Q510 needs — a characterisation of $\widehat{A}$ cannot be read off the A-direction, because the A-direction is blind to the renormalisation. Note the scope limit inherited from the previous paper's Theorem C: the family considered is *point-indexed and diagonal*, $\Psi_\chi(\mathbf{M}) = \prod_{m\in\mathbf{M}} \chi(w_m)$. Koncki's discrepancy at $N = 2$ is unipotent *off-diagonal in label space* (Rick, UID 824), so it lies outside this family and outside Corollary 22's reach.

5 The pre-registered branch, and what it cost me

The brief wrote the branch before the measurement, and I quote it so the grading is checkable:

> "**PRE-REGISTERED BRANCH, written before measuring.** If $\widetilde{A}$ has **no** A-type poles, then Task 2 is vacuous too and **gap 2 was never a gap**. Report that plainly and stop; do not dress a vacuity as a result. I expect $\widetilde{A}$ *does* have them (the bulk is a type-A-symmetric model and $\mathcal{E}^C$ is a boundary normaliser, so nothing cancels the A-type collisions), but I have been wrong about exactly this shape of expectation twice this week, both times on a normaliser."

The branch fired on the arm I did not expect, for the third time this week. $\widetilde{A}$ has no A-type poles — not in the basis (all 35 nonzero entries at $N \le 3$ are Laurent polynomials, §6), and not in the structure constants (the 13/115 non-Laurent baseline constants have *pair-divisor denominators only*; the A-type set is empty). So there was no A-type defect for Ψ to create or destroy, and **gap 2 read as a difference question — "does Ψ break the A-direction?" — is vacuous.** I say that plainly, as instructed.

Where my stated reason went wrong is worth naming precisely, because the error was not in the conclusion but in the *direction of the inference*. I wrote "the bulk is a type-A-symmetric model ... so nothing cancels the A-type collisions." The symmetry is real and I had the right object; I then used it to predict that the collisions *survive*. It is the very thing that makes them *harmless* — u-symmetry is exactly the hypothesis of Lemma 6. I had the mechanism and read its sign backwards.

The two readings of gap 2 therefore come apart, and only one of them is empty:

- *As a difference question* ("is $\widehat{A}$ worse than $\widetilde{A}$ here?"): vacuous, and provably so — Corollary 22 says no admissible renormalisation changes the A-type lattice at all.
- *As a regularity question* ("are the structure constants regular on $\{w_i = w_j\}$?"): **not vacuous**. It was unproved before today for either basis, at any N; `2026-10-17-c2` proved regularity only on the pair and boundary divisors, and its Theorem B's unit argument gives nothing here (a unit cannot cancel a pole in the basis collapse, but the structure constants *reshuffle*). Theorem 19 settles it for all N and for the whole family, and Theorem 14 says something strictly stronger than regularity: the span is not merely *contained in* a ring, it *equals* the GKM lattice.

The honest summary: the session's measurement was vacuous and its theorem was not. A vacuity has a mechanism, and in this case the mechanism (Remark 7) also explains the *non*-vacuous half of the previous paper — why the pair divisors, and only they, carried poles.

6 Verification

All computations are pure Python + SymPy (SageMath is not installed in this container) in `proofs/code-1018-atype/`. The engine `q484_engine.py` exists in four places in the tree (`code-1018-atype/`, `code-1017c2-coboundary/`, `code-1017-nonvanH/`, `reviews/scratch-20261017c2/`). Checked as a *set* at the start of this session: all four byte-identical, md5 `8b37ab1b`, so the stale-copy warning carried in the brief is discharged. Re-checked as a set after the locator fix described in (M3): all four `f53c0258`. I verify the *set*, not one pair, because a drift check run on one pair reports health for the copy it did not open.

V0. Root lengths, from first principles. Enumerating B_n and C_n for $n = 2, 3, 4$ and classifying by $|\alpha|^2$: the norms are $\{1, 2\}$ in B and $\{2, 4\}$ in C, so $e_a + e_b$ is *long* in B and *short* in C, while $h \cdot e_m$ is *short* in B and *long* in C. This is the measurement behind the nomenclature correction to `2026-10-17-c2` (Remark 2 there, commit `7002d00`).

V1. The divisor classifier. `divisors.py` decides a factor's divisor class *by vanishing* — substituting each candidate divisor's defining relation — rather than by matching the printed form, which would miss $1 - w_2/w_1$ and $(w_1 - w_2) \cdot (-1)$. Positive controls: 11 planted factors, each placed in the correct one of `A/pair/bdy/qdef/other/unit`, 11/11. Crucially $(w_a - q w_b)$ and $(1 - q w_a w_b)$ land in `qdef` and are never merged into a geometric class: $\{w_a = q w_b\}$ and $\{w_a = w_b\}$ differ as sets and coincide only at $q = 1$.

V2. The basis is Laurent (`basis_atype.py`). At $N = 2$ (10 nonzero entries) and $N = 3$ (35 nonzero entries), type C: *no non-monomial denominator occurs at all*, so in particular zero A-type poles. This is stronger than Lemma 5, which claims only ord ≥ 0 at A-type divisors; I do not claim the stronger statement as proved.

V3. Lemma A1, with three controls (`atype_gkm.py`). Exact rational w with $w_j = w_i$ imposed *exactly*, q kept symbolic, so each test is an identity of rational functions in q; two independent w-seeds.

	$N=2$	$N=3$	$N=4$
(i,j,**M**,**I**) instances, both seeds	8/8	96/96	768/768
of which trivial $0=0$	4	54	464
with a nonzero side	4	42	304
$\rho^A = 1$ and **I**-independent, every orbit	yes (2)	yes (12)	yes (48)
C1 off divisor ($w_j = \frac{5}{4} w_i$)	fails 6	fails 62	fails 442
C2 planted $\rho = 1 + q$	fails 4	fails 42	fails 304
C3 wrong partner $s_{ik}\mathbf{M}$, $k \neq i, j$	—	fails 62	fails 456

All three controls fire. Each is recorded with the factor it is sensitive to: C1 to the collision itself; C2 to the *value* of the multiplier; C3 to *which point* the congruence pairs **M** with. C2's "holds" column is 4, 54, 464 — *exactly* the trivial-$0=0$ counts, i.e. the plant was rejected on every one of the 4, 42, 304 nonzero cells and structurally spared only where $0 = \rho \cdot 0$ holds for any ρ. Reporting the nonzero column separately is what makes that green readable.

V4. The colength arithmetic (`diag_order.py`). $\mathrm{ord}_{(w_i - w_j)} \widetilde{A}_{\mathbf{I},\mathbf{I}}$ computed *directly from the engine* — the S_n symmetrisation — and compared against the formula derived from the root product (1). These are independent mechanisms; it was the same kind of cross-check that caught the $\mathcal{E}^B$ error on 2026-10-17.

	$\sum_{\mathbf{I}}$ ord vs. 2^{N-2}	per-label ord $= [i \notin \mathbf{I}, j \in \mathbf{I}]$	(M4) engine = root product
$N=2$, C, 1 pair	$1 = 1$	all 4 labels agree	4/4
$N=3$, C, 3 pairs	$2 = 2$	all 8 labels agree, every pair	8/8
$N=4$, C, 6 pairs	$4 = 4$	all 16 labels agree, every pair	16/16
$N=2$, B, 1 pair	$1 = 1$	all 4 labels agree	n/a
$N=3$, B, 3 pairs	$2 = 2$	all 8 labels agree, every pair	n/a

The type-B rows were run as a *measurement*, because the engine has no type-B `diag_rootproduct` and (1) must not be reused there (that reuse is precisely the $\mathcal{E}^B$ error). They have since become a check on Lemma 12, which proves the type-B count from Motegi's own `normalizationWB`; see Remark 24.

V5. The residue mechanism of Lemma 5 (`residue_pairing.py`). For five strata $(\mathbf{M}, \mathbf{I})$ with $\{1,2\} \subseteq \mathbf{M}$: (T1) the full $\sum_{\sigma \in S_n}$ is regular at $w_1 = w_2$ in 5/5; (T2) *every involution pair* $\{\sigma, \sigma'\}$ is *already* regular, 5/5 strata, and the involution is fixed-point-free as claimed. So the cancellation is pairwise, which is the mechanism the proof asserts and not a global accident.

The control — a partial sum over a σ-set *not* closed under the involution must have a pole — fires in 2 strata and is silent in 3. The three silences are **empty domains, with a named cause in each case**, not a blind instrument:

- $\mathbf{I} = \emptyset$ gives $r = 0$, and both ranges carrying a $\frac{qv - v'}{v - v'}$ factor are indexed by $k \leq r$ — so the stratum *structurally deletes every factor the control tests.* Measured: 0 of 6 summands have a pole.
- $\mathbf{I} = (2,3)$ and $\mathbf{I} = (1,2)$: the factors $\Lambda_j \ni (v_j - w_1)(v_j - w_2)$ supply numerator zeros on $\{w_1 = w_2\}$ that kill each summand's pole term-by-term. Measured: 0 of 6, resp. 0 of 2, summands have a pole.

In the two strata where the tested factor *is* present ($\mathbf{M} = (1,2,3)$, $\mathbf{I} = (1,)$: 4 of 6 summands polar; $\mathbf{M} = (1,2)$, $\mathbf{I} = (1,)$: 2 of 2), the control fires, ord $= -1$ both times.

V6. Structure-constant census, denominators classified (`sc_census.py`). Same triangular solve as the previous paper's `arm.py`, so the 13/115 and 0/115 baselines are reproducible, but every non-monomial denominator factor is classified. Reported as *sets* of triples (λ, μ, ν), never as counts. Pre-registered in `PREDICTIONS.md` before running: **zero A-type denominators in every arm**, including the four arms that produce 57/115 non-Laurent constants — a single $(w_a - w_b)$ among those 57 would refute Theorem 19.

		$N=2$ (complete)				$N=3$	
arm	$\chi(v)$	nz	non-Laur.	`A`	`pair/bdy`	non-Laur.	`A`
`id`	1	20	1	**0**	1/0	13	**0**
`psi`	$(1-qv^2)/(1-q)$	20	0	**0**	0/0	0	**0**
`x2psi`	$v^2(1-qv^2)/(1-q)$	20	0	**0**	0/0	not run	
`sq`	$\big((1-qv^2)/(1-q)\big)^2$	20	1	**0**	1/0	not run	
`h1`	$(1-qv)/(1-q)$	20	8	**0**	1/8	not run	
`unrm`	$1-qv^2$	20	8	**0**	1/8	not run	
`xpsi`	$v(1-qv^2)/(1-q)$	20	8	**0**	0/8	not run	
`x`	v	20	8	**0**	1/8	**57**	**0**
`atp`	*plant, see below*	20	14	**13**	1/8	RUNNING	

At $N = 2$ the sweep is complete over all eight admissible arms: 34 non-Laurent constants in total, and the `A` column is 0 in every one of them — 34 separate opportunities to refute Theorem 19, all declined.

At $N = 3$ the three arms that ran reproduce the previous paper's table exactly, and the hardest one is decisive. `id`: 115 nonzero, 13 non-Laurent, all 13 `pair`, `A`-set $\emptyset$. `psi`: 0/115. `x`: **57** non-Laurent, of which 13 `pair` and 57 `bdy` — and `A`-set $\emptyset$. `x` is the arm that fails *both* (COB) and (SR), so it is where a third condition would have shown up if there were one: 57 **separate opportunities to refute Theorem 19 at $N = 3$, all declined.**

The cap. The five arms marked "not run" at $N = 3$ were killed to hold the machine at its four-core load, and `atp` at $N = 3$ had not finished. I record the cap rather than present the table as a complete sweep; the $N = 3$ coverage is three arms of nine, chosen as baseline, the predicted-clean case, and the most aggressive failing case. **The plant fires.** Arm `atp` divides $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ by $(w_1 - w_2)$ for $|\mathbf{M}|$ odd. This is *not* a point-product renormalisation and it deliberately violates the unit hypothesis of Lemma 17, so it must produce A-type poles or the **0** column above is decoration. It produces 13 of them at $N = 2$, every one classified `A` with the correct index pair $(1,2)$ and the correct factor $w_1 - w_2$, on the *same* solve and the *same* classifier that return 0 for all eight admissible arms. That is the two-arm structure the **0** column needs: the instrument is demonstrably able to see the thing it is reporting absent.

The `atp` plant probes one hypothesis only — it introduces a *pole*, so it tests "Ψ regular" and nothing else. The remaining hypotheses are lesioned separately in V7, which is where the sharp controls are.

V7. Lesioning the theorem's own hypotheses (`lesion.py`). Theorem 19 rests on exactly three things; each is lesioned once. A lesion that *survives* is a stronger theorem for free; one that *breaks* shows the hypothesis is load-bearing rather than decorative. All at $N = 2$, type C; baseline 20 nonzero, 1 non-Laurent, `A`-set empty.

L1 — is $\rho^A = 1$ needed, or would any constant do? Algebraic, no computation: if the multiplier were a constant unit c, so that $\mathcal{G}_c = \{f : f(\mathbf{M}) \equiv c\, f(s\mathbf{M})\}$, then $(fg)(\mathbf{M}) \equiv c^2 (fg)(s\mathbf{M})$ — weight $c^2 \neq c$ unless $c = 1$, so $\mathcal{G}_c$ is *not* closed under multiplication. $\rho^A = 1$ is necessary, and this is literally the $\rho^2 - \rho$ mechanism of Remark 7.

L2 — the colength match. Multiply *one* basis column $\phi_{\mathbf{I}_0}$ by $(w_1 - w_2)$. This keeps every entry regular (so it does not touch Lemma 5), is *label*-indexed, and raises $\sum_{\mathbf{I}} \operatorname{ord} \widetilde{A}_{\mathbf{I},\mathbf{I}}$ from 2^{N-2} to $2^{N-2} + 1$, making $\mathcal{L}$ a colength-1 *proper* sublattice of $\mathcal{G}$. Measured:

$\mathbf{I}_0$	non-Laurent	A-type	
$\emptyset$	1	0	structurally spared — see below
$\{1\}$	2	1	FIRES
$\{2\}$	4	3	FIRES
$\{1,2\}$	6	6	FIRES

So the colength match is load-bearing: break it by one and A-type poles appear. The $\mathbf{I}_0 = \emptyset$ row is spared for a nameable reason, not because the control is blind: $\emptyset$ is minimal in Motegi's order, so $\widetilde{A}_{\emptyset,\lambda} = 0$ for every $\lambda \neq \emptyset$ by (M3); hence $c^{\emptyset}_{\lambda\mu}$ is either zero or carries a compensating factor of $(w_1 - w_2)$ from the numerator. Triangularity deletes the lesion exactly on the one column where it would have to act alone.

L3 — point-indexedness, and what the hypothesis really is. Two weights, both units at every point, neither a point-indexed product:

Ψ	in $\mathcal{G}^\times$?	non-Laurent	A-type	
$1 + (w_1 - w_2)$ on $\mathbf{M} = \{1\}$, 1 else	yes ($\equiv 1$)	5	**0**	SURVIVES
2 on $\mathbf{M} = \{1\}$, 1 else	**no** ($2 \not\equiv 1$)	7	**6**	FIRES

This pair is the reason Theorem 19 is stated for $\mathcal{G}^\times$ rather than for the point-product family: the first lesion survives, so point-indexedness was never the hypothesis, and the second fires, so pointwise invertibility is not enough either. The two weights differ *only* in whether the perturbation is $\equiv 1$ modulo the uniformiser. Note also that the first lesion's five non-Laurent constants are all `bdy` — it does disturb the *boundary* divisors, as it must, since nothing here constrains those; it is the A-type column that stays empty.

V8. Theorem 14's *conclusion*, tested directly (`lattice_equality.py`). V4 checks the two colengths, which is the *proof's* arithmetic; this checks the equality $\mathcal{L}_{ij} = \mathcal{G}_{ij}$ itself. The nontrivial inclusion is $\mathcal{G} \subseteq \mathcal{L}$: take a generator of $\mathcal{G}_{ij}$ over $\mathcal{O}_{ij}$ — the s_{ij}-symmetric orbit sums $\mathbf{1}_{\mathbf{M}} + \mathbf{1}_{s\mathbf{M}}$, the elements $t \cdot \mathbf{1}_{\mathbf{M}}$, and $\mathbf{1}_{\mathbf{M}}$ for s_{ij}-fixed $\mathbf{M}$ — expand it in the ϕ basis by the *same* triangular solve the structure constants use, and require every coefficient to have ord ≥ 0. At $N = 2$: 4/4 generators clean in type C and 4/4 in type B (an independent corroboration of Lemma 12).

The control is $\mathbf{1}_{\mathbf{M}}$ for a *moved* $\mathbf{M}$, which is *not* in $\mathcal{G}$ ($1 - 0 = 1 \notin t\mathcal{O}$); Theorem 14 then forces some coefficient to have ord $= -1$. It FIRES, 1/1 in both types. Without it the clean column would only say the solve returns regular coefficients for *everything* I happened to feed it. The $N = 3$ arm was still running when this was written.

7 Gaps and questions

Remark 24 (the type-B gap I expected to file, and did not). I had this paragraph drafted as a Gap: Lemmas 4, 5 and 6 are type-free, but Lemma 11 was proved from the type-C root product

(1), and for type B I had only the *measurement* of V4 ($\sum_{\mathbf{I}} \mathrm{ord} = 2^{N-2}$ and the per-label formula, at $N = 2, 3$ only, because the engine has no type-B diagonal root product). Reading L1307–1340 closed it: Lemma 12. Theorems 14 and 19 are therefore *proved* for both types, and the type-B rows of V4 are now a check on a proof rather than its only support. Q514, posed at the start of this session for exactly this gap, is answered within it and retired.

Gap 1 ($N \geq 5$ is not measured)**.** *The symbolic census is run at $N \leq 4$ (Lemma A1) and $N \leq 3$ (structure constants). The* proofs *are uniform in N, so this is not a gap in the argument; I record it because the colength match $2^{N-2} = 2^{N-2}$ is the one step where an off-by-one in the orbit count would be invisible at small N — at $N = 2$ both sides are 1, and 1 is the value a count of almost anything can take. $N = 4$ is the first value at which $2^{N-2} = 4$ is distinguishable from the alternatives $N - 1 = 3$ and $2^{N-2} + 1 = 5$, and both halves of the colength match are measured there: Lemma A1 at 768 instances over 48 orbits (V3), and $\sum_{\mathbf{I}} \mathrm{ord}\, \widetilde{A}_{\mathbf{I},\mathbf{I}} = 4$ for all six pairs with the per-label formula holding on all 16 labels (V4). At $N = 2$ both sides are 1, and 1 is the value a count of almost anything can take.*

Gap 2 (what this does *not* say about $\widehat{A}$ versus Koncki's class)**.** *Corollary 22 is a statement about point-indexed* diagonal *renormalisations. Rick's UID 824 result is that Koncki's single-entry discrepancy at $N = 2$ is* unipotent off-diagonal in label space, $\mathrm{Koncki}_{\emptyset} = \widetilde{A}_{\emptyset} + c\,\widetilde{A}_{\{1,2\}}$ *with $c = q(1-q)/(1-1/(w_1 w_2))$ having a simple pole on the pair divisor H_{12}. That lies outside the family here and outside the previous paper's Theorem C. The A-type direction cannot contribute to the Q510 identification at all, precisely because it is blind to the whole admissible family.*

New questions. **Q513** — does "spans exactly the GKM lattice at all three root families" pin $\widehat{A}$ up to a point-product unit? Theorem 14 supplies the A-type lattice equality and the previous paper's Theorem B supplies the pair-divisor colength argument. If the three equalities plus (M3) plus (1) determine the basis up to Ψ_χ with χ a unit at every divisor, then Q510 becomes a theorem about characterisations rather than a table comparison. Gate: the A-type direction imposes no condition, so count the remaining freedom before claiming a characterisation. **Q514** — *answered and retired within this session* (Remark 24): `normalizationWB` (L1386) is indeed a B_N positive-root product, its A-type factors are again $\prod_{j \in \mathbf{I}^c} \prod_{l: i_l > j} (1 - w_{i_l}^{-1} w_j)$, and the whole type-dependence of the diagonal is the boundary root $2t_k$ versus t_k. See Lemma 12.

Acknowledgement

Rick's Lemma QC (UID 819) and his independent GKM check of $\widehat{A}$ at $N = 2$ (UID 824, run on *his* instrument, which is what makes it corroboration rather than my own evidence counted twice) set up the question this paper answers. He explicitly declined $N \geq 3$; that was this session's target and Theorem 19 covers all N.